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Topologies of multiterminal HVDC-VSC transmission for large offshore wind farms

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ABSTRACT

Topologies of multiterminal HVDC-VSC transmission systems for large offshore wind farms are investigated. System requirements for multiterminal HVDC are described, particularly the maximum power loss allowed in the event of a fault. Alternative control schemes and HVDC circuit topologies are reviewed, including the need for HVDC circuit breakers. Various topologies are analyzed and compared according to a number of criteria: number and capacity of HVDC circuits, number of HVDC circuit breakers, maximum power loss, flexibility, redundancy, lines utilization, need for offshore switching platforms and fast communications.

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1. Introduction

HVDC transmission is presently being considered for large offshore wind farms located far away from the land based grid. Current Source or Line Commutated HVDC systems have been used extensively on land in the past [1] but for offshore wind farms voltage source HVDC is now considered to be more attractive [2].

For wind farms, an HVDC system is formed by a rectifier connected to the wind farm, a DC cable circuit and an inverter connected to the land based grid. The rectifier and inverter can either use thyristors in a line commutated converter (LCC) or IGBTs in a voltage source converter (VSC). A number of authors have discussed the potential for HVDC transmission from offshore wind farms [2,3] and compared HVDC and HVAC to assess economic viability [4,5].

Several studies have been undertaken on Doubly Fed Induction Generators (DFIG) based offshore wind farms with HVDC links (either VSC or LCC). HVDC-LCC with a wind farm based on DFIGs has been investigated by Refs. [6,7]. The wind farm LCC needs a voltage source, typically either a STATCOM [6] or from the DFIGs [7]. The coordinated control of DFIG and HVDC-LCC was investigated in Ref. [7]. VSCs can provide independent control of active and reactive power and there have been important advances in HVDC-VSC converters in the last decades [8,9].

Xu et al. [10] simulated the use of HVDC-VSC systems, using a three level neutral point clamped converter, for connecting DFIG-based wind farms. The grid side VSC was responsible for controlling the DC voltage and the AC grid voltage, while the wind farm side VSC controlled the frequency and voltage of the wind farm. Xu and Anderson [11] presented case studies for wind farms connected with HVDC, comparing HVDC-VSC and HVDC-LCC systems taking into account black start, behaviour during land based grid faults and behaviour during wind farm network faults. Fault ride-through of DFIGs connected through HVDC-VSC systems has been considered in Ref. [12], where two methods to achieve fault ride-through are presented, one with full communication and power reduction during the fault and the other without needing communications between the HVDC-VSC and the wind turbines. The combination of HVDC-VSC and squirrel cage induction generators was studied in Ref. [13] focusing on fault ride through capability enhancement. Simulations of squirrel cage based wind farms connected through VSC-HVDC are also reported in Ref. [14].

Multiterminal HVDC systems [15] have been discussed since 1963 [16] when the first parallel multiterminal HVDC system was proposed. A series multiterminal system was discussed in 1965 [17]. The usual principle of operation for voltage source converters, which are connected in parallel, is that one converter controls the DC voltage while the others regulate their power transfer. Lu and Ooi [18] addressed the optimal collection of wind power by multiterminal HVDC-VSC. They described a HVDC control structure

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where each HVDC-VSC controlled the AC voltage and the power injected by the corresponding wind turbines. The HVDC-VSC connected to the main grid was responsible for controlling the DC voltage. Current source converters for wind farm off-shore systems were considered in Ref. [19].

This paper considers the circuit topology of multiterminal HVDC-VSC transmission systems for large offshore wind farms and their integration into the land based AC grid, with particular emphasis on loss of power in-feed in the event of faults. The main paper contribution is the comparison of the different circuit topologies and the description of their operation under different system faults. Different topologies using the minimum number of HVDC circuit breakers are considered in order to determine if they can deal with the grid code requirements at reasonable cost. The control needed to ensure the required performance under fault conditions is discussed.

2. Multiterminal HVDC for offshore wind farms

Multiterminal HVDC-VSC systems are composed of a number of different converters which are connected to a common HVDC circuit. The wind turbines could be squirrel cage induction generators (SCIG), which are directly connected to the AC side of the HVDC-VSC, Doubly Fed Induction Generators (DFIG) or Full Converter Generator (FCG). These alternatives are illustrated in Fig. 1, where two SCIG based wind farms are connected to the HVDC-VSC of WF1 and WF4, a DFIG wind farm is connected to the HVDC-VSC of WF3 and a FCG based wind farm is connected to the HVDC-VSC of WF2. Each offshore HVDC-VSC unit needs an offshore platform on which the converter is installed . The number of connections to the main grid can vary depending on the application. A configuration with four different connections to the main grid SS5, SS6, SS7 and SS8 is illustrated in Fig. 1.

2.1. HVDC circuit requirements

The potential of offshore wind farms has led to the inclusion of this form of renewable energy in the energy planning of a number of countries [20–22]. A preliminary design of such a system depends on technical-economic factors and constraints imposed by the grid utility to which the system will be connected. The economics of the system is governed by the circuit lengths, ratings of the circuits and converters, the number of fast HVDC circuit breakers and isolators and the need for offshore platforms and fast communications. Further, there are a number of technical issues which are to be considered: the system flexibility, short and long term failure operation, redundancy and efficient use of the installed lines.

In the UK, the Great Britain Security and Quality of Supply Standard (GB SQSS) Expert Group has recommended criteria for offshore networks connecting to the onshore grid [23]. According to these recommendations the multiterminal HVDC system has to assure the following:

- The DC voltage has to be controlled both in normal and faulted operation.
- In the event of a fault in the land based AC grid, the system has to be able to provide the support to the AC grid specified by the grid code.
- In case of failure in any grid or converter (multiple simultaneous failures are not considered) the system has to assure that there is not a change in the power supplied to the main AC grid larger than $P_{\text{max-fail}}$. (e.g. 1320 MW in Great Britain [24]). This requirement will be referred to as 'the maximum power loss criterion' in the subsequent sections.

The maximum power loss criterion is specially important from the system operator point of view, since it can guarantee that there will not be a loss of power infeed larger than the specified value and therefore the system stability can be kept in the event of any single fault.

2.2. HVDC switchgear

The absence of cyclic moments of current zero in a DC system inherently makes DC current switching more difficult than for AC systems, as arcs require a current zero in order to extinguish. There are two main types of HVDC circuit breaker: electromechanical and solid-state.

There have been many electromechanical HVDC circuit breakers developed over the past 5 decades. These can be grouped into three main interruption techniques: (1) inverse voltage generating method, (2) Divergent Current Oscillating method, and (3) inverse current injecting method [25]. Of these, the inverse-current injecting method is suitable for higher voltage and current ratings. This creates a current zero by superimposing a high frequency inverse current on the DC current by discharging a pre-charged capacitor through an inductor. A conventional AC circuit breaker can be selectively utilised in this method and the cost of components required for such an electromechanical DC circuit breaker would not be significantly higher than that of an AC circuit breaker. Typical applications of electromechanical HVDC circuit breakers can be found in the neutral bus of bipolar HVDC-LCC schemes [26]. Electromechanical HVDC circuit breakers are available up to 500 kV, 5 kA and have a fault-clearing time of the order of 100 ms [25,27].

There are some cases where a clearing time which is much faster than possible with electromechanical HVDC circuit breakers is required. Example applications include pulsed power [28] and traction [29]. Solid-state circuit breakers are suitable and able to interrupt current within a couple of milliseconds [28,30]. They are generally based on Integrated Gate Commutated Thyristors (IGCT) as they have lower on-state losses than IGBTs [31]. Current flows through the IGCT during on-state operation. In order to interrupt the current, the IGCTs are turned off and the voltage quickly increases until a varistor starts to conduct. The varistor is designed to block a voltage above the system voltage level. The main disadvantages of the solid-state circuit breakers are their high on-state losses and capital costs. Typical ratings of solid-state circuit breakers in operation are 4 kV, 2 kA [28,29], although in [30] ratings of up to 150 kV, 2 kA were considered.

In a multiterminal HVDC system without HVDC circuit breakers the whole system voltage would have to be brought to zero in order to clear a fault. In the case of multiterminal HVDC-LCC systems, the overcurrent can be limited by its DC current control function and the fault can be cleared by the action of thyristor valve control and protection. After a de-ionisation period, the system can be restarted in 100–300 ms [32,33]. In the case of multiterminal HVDC-VSC systems, such a fast re-start would be impossible without the use of HVDC circuit breakers. If a fault occurred on a multiterminal HVDC-VSC system, the free-wheeling diodes used in the VSC would cause DC current to continue to flow into the fault even if the IGBTs were blocked. If HVDC circuit breakers are not used, it would be necessary to open the AC circuit breakers at all converter terminals. It would take a relatively long time to restore the system since it would be necessary to charge the DC system to rated voltage before restarting.

In the case of a large multiterminal HVDC-VSC system it would not be feasible to allow the whole system voltage to go to zero in the event of a fault, as this would result in a significant loss of power. DC circuit breakers which can clear the fault within a few milliseconds are required [34]. This is to meet the requirements of preventing the fault current rising very fast towards very high values due to

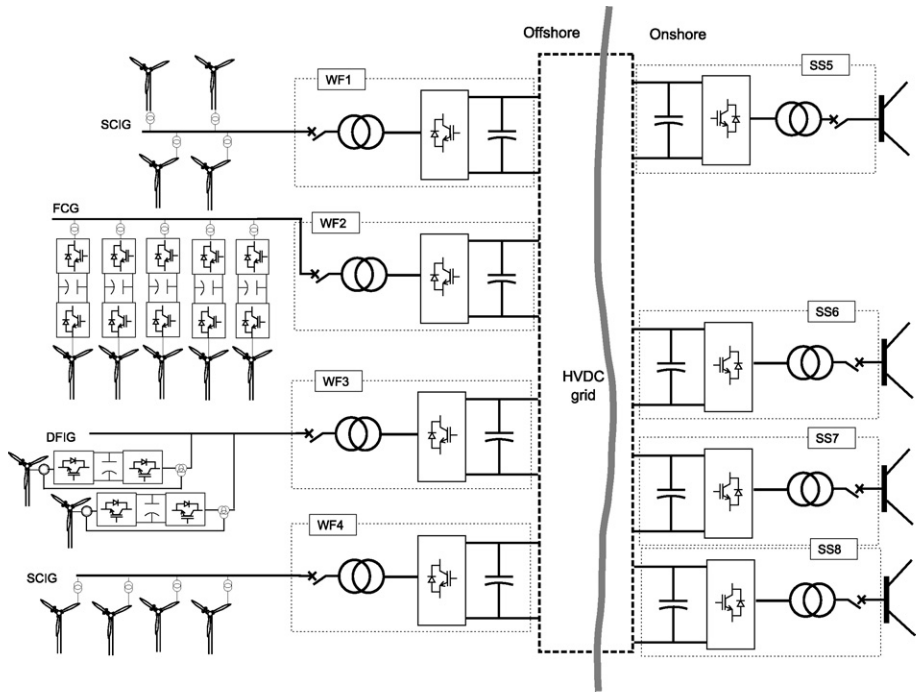

Fig. 1. Multiterminal HVDC based on VSC scheme.

the low system impedance and preventing voltage collapse of the whole HVDC system. A solid-state HVDC circuit breaker would be necessary since the operating time of an electromagnetic HVDC circuit breaker is too slow for this requirement. It is sensible to minimise the number of these solid-state HVDC circuit breakers in the system due to their very high costs and on-state losses. Faults are also rare on DC cables and on the DC side of converters, so investment in solid-state HVDC circuit breakers at the extremities of all circuits cannot be justified.

A sectionalisation technique which uses an optimised number of solid-state HVDC circuit breakers will be used in this paper. This means that the overall system can be separated into two subsystems, of which the faulted subsystem has a power generation capacity no greater than $P_{\text{max-fail}}$ to the main land based grid. Throughout the rest of this paper, the term HVDC circuit breaker will refer to the solid-state type.

3. Multiterminal HVDC control

Multiterminal HVDC control is strongly linked with the system topology. The behavior of the overall system both in normal and fault conditions is determined by the control system. Therefore, although the present work does not delve into the details of the control system, a general approach is given focusing specially on the system behavior under faults. Each of the converters of the multiterminal HVDC system [18,35] will control the reactive power exchanged with the connected AC network and the active power injected or extracted from the HVDC grid. However, depending on the kind of AC grid connection, different control modes can be defined: land based AC grid mode and wind farm grid mode.

3.1. Land based AC grid control mode

The HVDC-VSCs connected to the main land based grid control the DC voltage and exchange real power with the land based AC grid. Two different control approaches may be considered:

- Master–slave with communications. One of the HVDC-VSC units is responsible for controlling the DC system voltage. The other VSCs follow a given power reference point which can be constant or assigned by a master converter. In the event of a failure of the main converter, another converter takes over the HVDC voltage control. The communications system ensures that only one converter acts as the master. Communications can also facilitate power sharing between nodes, since the master converter can give power references to the slaves.
- Coordinated control. The VSCs control the HVDC grid voltage in a coordinated manner.
- Without communications: this uses a droop-based technique [36], where each converter has a defined linear relationship between HVDC voltage and extracted power. Although this simple control scheme has been shown to provide good performance in some situations, further investigations are required to examine how it can deal with voltage variations between nodes due to DC circuit resistances and how it will behave in the event of system reconfigurations or converter failure.
- With communications: complex coordinated systems can be implemented with the use of communications, taking into account the different voltages and currents in the HVDC grid. It is also possible to employ a hybrid approach that can provide effective operation when the communication system is in operation and assure safe operation when the communications fail.

Each HVDC-VSC can control the reactive power injected or extracted from the main grid, within its rating.

3.2. Wind farm grid control mode

The HVDC-VSC controls the wind farm AC network voltage and frequency, and exports active power to the HVDC grid. The wind turbines can be controlled independently by their own converters (DFIG, FCG) or connected directly (SCIG) to the AC system created by the HVDC-VSC.

3.3. Multiterminal HVDC operation during AC grid faults

A grid-fault in one of the land based AC networks will limit (or eliminate) the power transfer from the adjacent VSC.

If the system has been devised as master–slave, when there is a fault close to the master VSC, another converter has to take over and control the DC voltage. This can be done by measuring the DC voltage or by means of communications. In any case, the system has to be fast enough to limit the DC voltage rise. If the fault is not close to the master VSC but on another land based AC circuit the problem can be addressed more easily, since the master VSC will continue controlling the DC voltage. Coordinated control systems can deal with individual converter failures more easily. They will adjust the power injected to the grid to the new situation as long as their current limits are not violated.

For faults in the land based AC grid, the export capacity of the overall system is reduced. In such cases, the priority is to control the DC voltage at the cost of losing power from the wind, either by dissipating power in chopper resistors, storing part of the energy as kinetic energy in the turbines, or reducing the mechanical torque of wind turbines [10,14,36]. Another DC voltage-current droop control in the wind farm VSC can be used to coordinate the power reduction from each wind farm [36].

4. Multiterminal HVDC topology analysis

Different topologies that can be used to transport power from very large offshore wind farms by means of multiterminal HVDC transmission systems are considered. Some of these topologies are well known in AC systems [37]. Operation under converter and DC faults is discussed since it is strongly linked to the choice of HVDC grid topology.

4.1. Point to point topology

Point to point topology (PPT) is shown in Fig. 2. It is based on multiple point to point links.

In the event of a converter or HVDC circuit failure, the appropriate action is to disconnect the faulted line by opening the AC circuit breakers of the grid side converter and let the turbines trip off on over-speed or DC-link over-voltages. The maximum power loss criterion is met without the need for HVDC circuit breakers and with $N_L(=N_{WF}=N_{SS})$ power lines rated for a single substation power. If there is a fault on a line, the wind farm connected by it will be lost. Therefore this configuration lacks flexibility.

4.2. General ring topology

The general ring topology (GRT) is shown in Fig. 3. The general ring topology is a multiterminal HVDC system with the lines connected to all the nodes composing a ring and therefore there are some lines which have to deliver the whole power of the system when the ring is opened. The ring can be operated in closed loop, having all the circuit breakers and isolators closed in normal operation, or in open loop, having a circuit breaker or an isolator in the ring opened.

In the case of a converter or DC grid fault, the first action is to open the two circuit breakers connected to the fault, leaving the system in open ring. When the fault current reaches zero, the isolators can isolate the faulted area and the appropriate circuit breakers can be connected again. An example of this operation is shown in Fig. 4:

1. Under normal operation, a closed ring as shown in Fig. 4(a).
2. At t_0, fault in L12 (Fig. 4(b)).
3. At t_1, the ring is opened, losing WF1 (Fig. 4(c)).

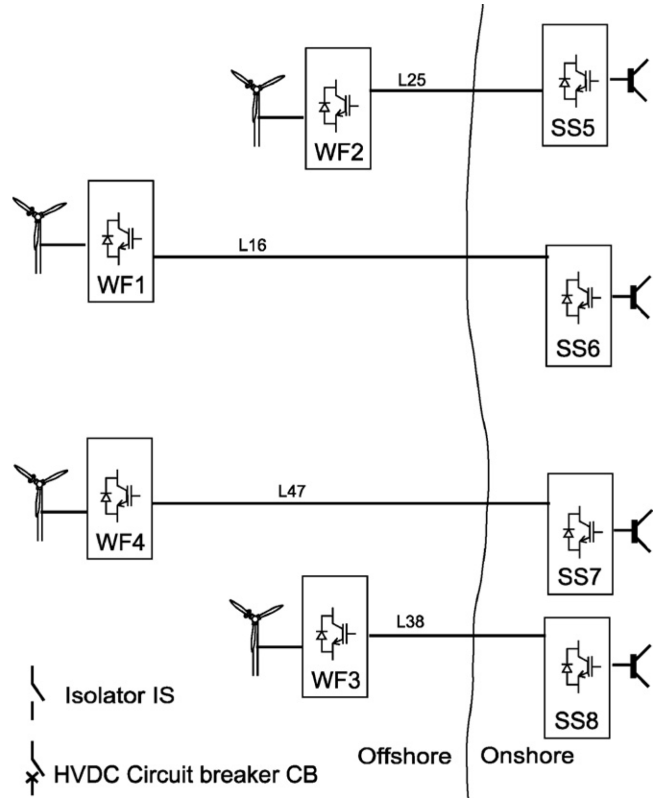

Fig. 2. Point to point topology.

4. At t_2 When there is no current in L12, IS12 opens (Fig. 4(d)).
5. At t_3 CB14 closes connecting WF1 (Fig. 4(e)).

In this final situation, L38 is loaded to $P_{WF1}+P_{WF3}+P_{WF4}$. The most critical case would be a fault in L38 or L25, which would imply that one of the circuits would have to transport the whole system power $\sum W_{WFi}$.

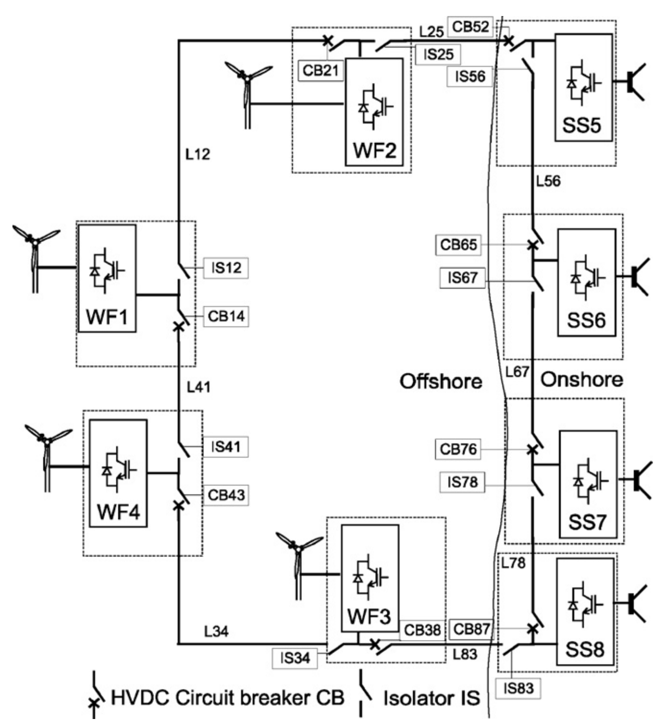

Fig. 3. General ring topology.

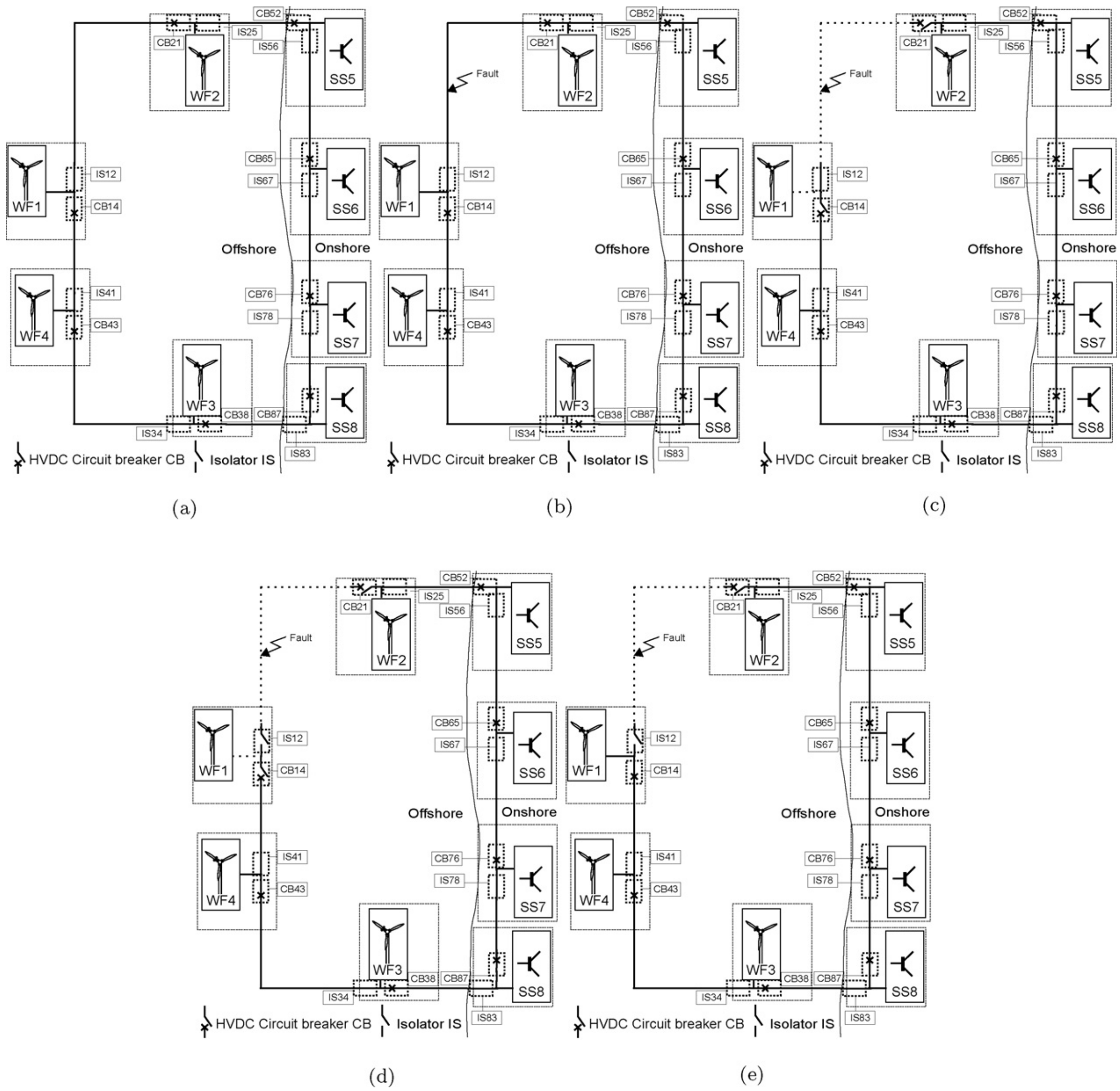

Fig. 4. General ring topology operation under a line fault. (a) General ring topology normal operation, (b) t_0 fault in the line L12, (c) t_1 circuit breakers CB14 and CB21 open, (d) t_2 isolator IS12 opens, (e) t_3 circuit breaker CB14 closes reestablishing the connection of the wind farm WF1.

The maximum power loss criterion can be fulfilled using $N_{cb}(=N_{WF}+N_{SS})$ HVDC circuit breakers, and $N_{WF}+N_{SS}$ lines (two of these lines connect offshore to onshore) whose maximum capacity corresponds to the total power generated. This configuration has flexibility at the cost of needing some lines of full power rating. For long term HVDC faults or maintenance, the system can be operated in open ring. Fast communications are needed to coordinate the circuit breakers and disconnect only the faulted circuits.

It is clear that by replacing the isolators with solid state HVDC circuit breakers, the supply from WF1 will not be lost at any moment but the cost will be increased.

4.3. Star topology

The star topology (ST), shown in Fig. 5, is a multiterminal HVDC system where each line that is connected to a wind farm or a substation is connected to a central star node. In this topology, the rating of each line corresponds to the rating of the wind farm or substation. The main drawback of this configuration is that a fault at the central node can cause the entire system to go off line. Therefore, in spite of the various advantages offered by ST, it is not a feasible topology for multiterminal wind connections.

Disregarding a fault on the central node, DC and converter faults can be handled by disconnecting the corresponding line using the HVDC circuit breakers. This configuration needs $N_{cb}(=N_{WF}+N_{SS})$ HVDC circuit breakers and $N_L(=N_{WF}+N_{SS})$ lines which are rated to the VSC connected. The star topology needs an offshore platform (or submarine installation) at the central node on which to place all the circuit breakers and star-point connections.

The flexibility of this configuration is not as good as GRT because for a permanent fault in a line from the central node to a wind farm, the full wind farm is permanently lost.

4.4. Star with a central switching ring topology

The two topologies GRT and ST can be combined to form a hybrid topology, star with a central switching ring topology (SGRT) (Fig. 6), which is essentially a star configuration with a central switching

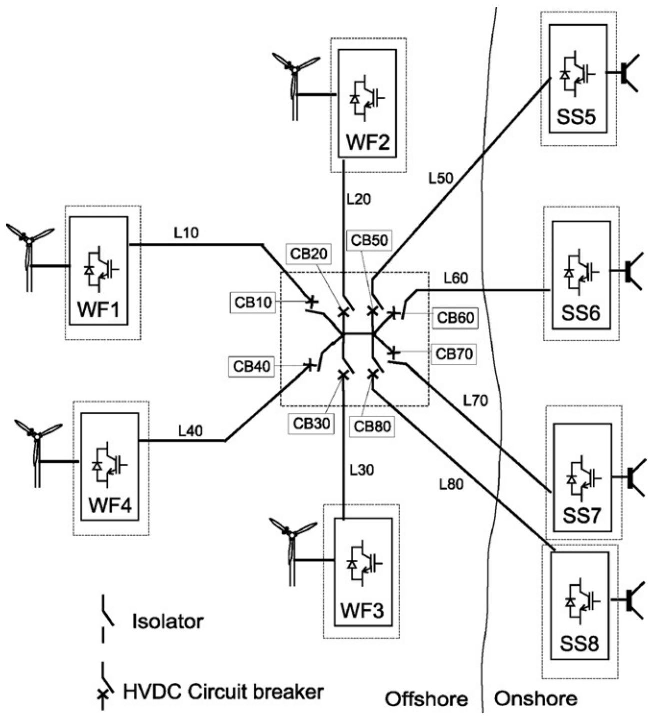

Fig. 5. Star topology.

ring. This configuration offers the advantages of GRT and ST: it needs only full power rating in the central ring, circuit lengths are kept to a minimum (wind farms or substations to central ring) and it can isolate a fault while meeting the maximum loss of power criterion. One of the main drawbacks of this configuration is the need for an offshore platform on which to install all circuit breakers in a ring.

The maximum power loss criterion can be fulfilled using $N_{cb}(=N_{WF}+N_{SS})$ HVDC circuit breakers and $N_L(=N_{WF}+N_{SS})$ star power lines rated to the wind farm or substation connected. The

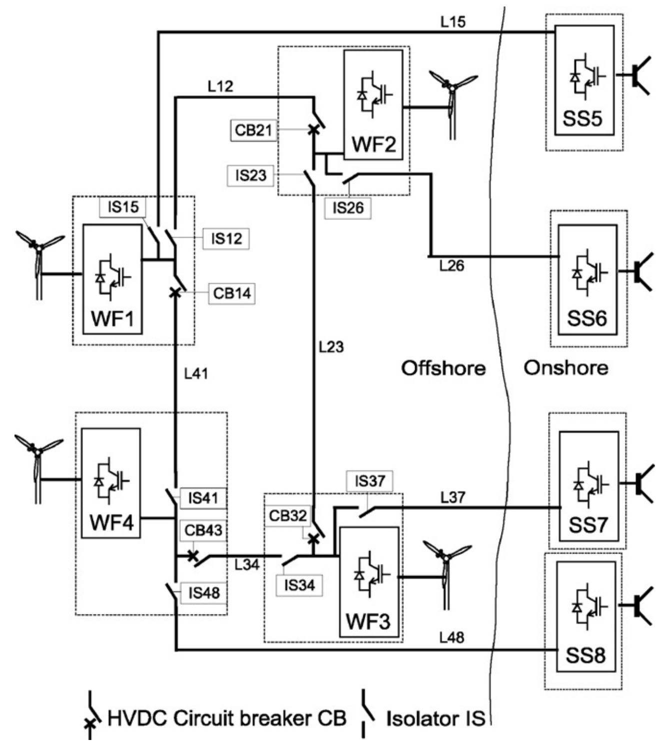

Fig. 7. Wind farms ring topology.

capacity of the central ring lines should be equal to the total system power. SGRT shows flexibility for short and long term HVDC faults. Regarding flexibility, SGRT has the same problem as ST: a full wind farm is lost for a permanent fault in a line from the central node to a wind farm.

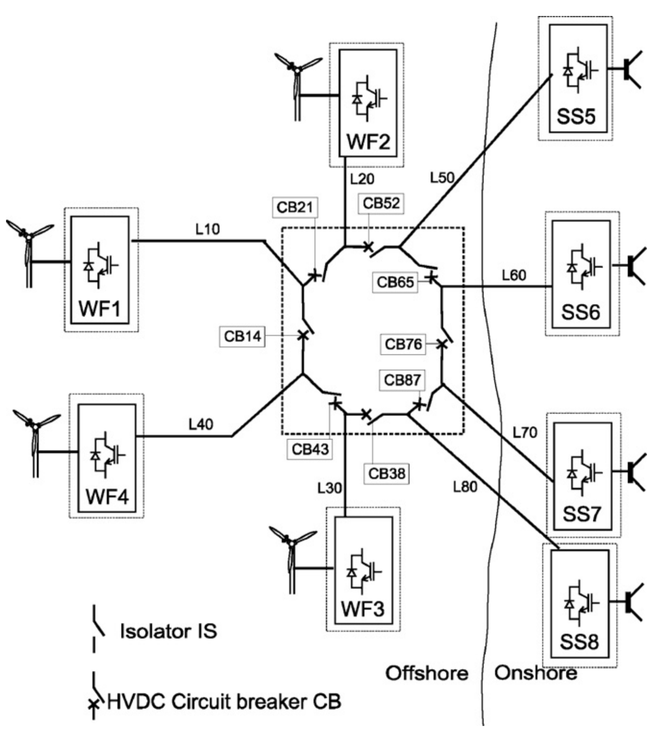

Fig. 6. Star with a central switching ring topology.

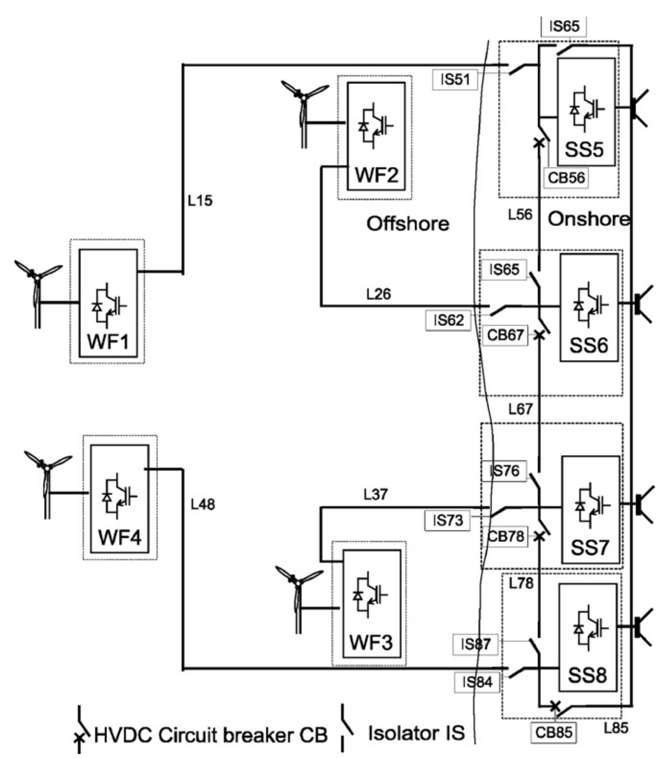

Fig. 8. Substations ring topology SSRT.

4.5. Wind farms ring topology

An interesting configuration that minimises the number of HVDC circuit breakers is the wind farm ring topology (WFRT), shown in Fig. 7. This topology comprises a wind farm ring having a number of HVDC circuit breakers equal to the number of wind farms and circuits connected to the substations. The topology is similar to the PPT but with the added flexibility of controlling the power flow between wind farms and land based substations. In the event of a critical fault, the faulted line can be isolated as if it were a PPT thus temporally disconnecting a wind farm. However the disconnected wind farm can be reconnected to another substation. Depending on distance and costs, the WFRT can be configured as a large ring or a concentrated switching ring as in the case of SGRT.

In the event of a circuit fault, the two HVDC circuit breakers in either side of the faulty location are opened thus disconnecting the wind farm and the substation. Once there is no current in the disconnected line, the line isolator can be opened and the HVDC circuit breakers can be reconnected isolating the faulted line. An example sequence is shown in Fig. 9:

1. Normal operation in closed ring (Fig. 9(a)).
2. At t_0, fault in L37 (Fig. 9(b)).
3. At t_1 the ring is opened (Fig. 9(c)), loosing WF3.
4. At t_2 when there is no current in the faulted line , IS37 opens (Fig. 9(d)).
5. At t_3 CB32 and CB43 close, connecting WF3 and re-establishing the wind farm ring (Fig. 9(e)).

The maximum power loss criterion can be met using $N_{cb}(=N_{WF})$ HVDC circuit breakers and $N_L(=N_{WF}+N_{SS})$ lines (N_{SS} lines offshore to onshore). The substation circuit capacity will be the rated substation power. For the wind farm ring, the line power between two wind farms can be the sum of the rated power of the two wind farms. WFRT features excellent flexibility as it can deal with faults in any line without permanently losing power from the wind farms. However it requires fast communications to coordinate the HVDC protections.

4.6. Substations ring topology

In a similar manner to the WFRT, a substation ring topology (SSRT) can be devised as illustrated in Fig. 8. This configuration performs similarly to WFRT with the difference that when there is a fault in a HVDC line, the isolated converter will be the wind farm HVDC-VSC (instead of the main grid HVDC-VSC in WFRT). This allows more flexibility in the main grid side at the cost of losing flexibility in the wind farm side both in faulty operation and during long term failures and maintenance. In most applications it will be more convenient to use WFRT, since faults in the HVDC circuits will allow continued extraction of all the power from the wind farms.

4.7. Other configurations

More complex solutions can be adopted at the cost of adding more lines and HVDC circuit breakers. Examples include mixed ring with HVAC and HVDC, double ring topology (WFRT plus SSRT), double whole system rings (doubling the GRT), double star topology (with different central nodes) and star plus ring topology (superposing a star and a ring topology). Such configurations can provide higher performance in certain cases at higher cost.

5. Comparison between topologies

5.1. General remarks

Regarding the number of HVDC circuit breakers, PPT does not need them. GRT, ST and SGRT need $N_{wf}+N_{ss}$ HVDC circuits breakers (one for each HVDC-VSC converter unit), while the partial ring topologies WFRT and SSRT need only N_{wf} or N_{ss} circuit breakers respectively.

Circuits of PPT, ST and SGRT can be rated at the same power of the converter where they are connected. The lines of GRT circuits have to be rated at full system power, thus are economically not attractive. WFRT and SSRT circuits can be rated to the rated power of a single station for the offshore to onshore circuits and to the double of the rated power of a station for the ring circuits. All the proposed topologies are multiterminal, except PPT. Multiterminal systems can be considered flexible since they allow redirecting the power flow in the event of a fault. The only exception is a fault in the central node with ST, where the maximum power loss criterion is not met.

In the event of a long term failure or a circuit disconnection, flexibility becomes crucial. With PPT the single wind farm is lost permanently. For GRT, if the fault is in the ring, the faulted circuit can be disconnected and remaining lines and converters can be in operation. For faults in the converter, the opened ring can still be in service. For ST and SGRT only the faulted line can be disconnected resulting in a converter lost. For WFRT a substation will be disconnected for faults in the offshore to onshore line, and the opened ring can be in service for faults in the ring. Similarly, for SSRT a wind farm will be disconnected for faults in the offshore to onshore line, and the opened ring can be in service for faults in the ring.

The topologies which employ a central switching node or ring (ST, SGRT) need an offshore platform for it. The others can be implemented without it. GRT, WFRT and SSRT need fast communication in order to coordinate the protections.

The most relevant features of each topology are summarized in Table 1.

5.2. Case study

A simplified case study is presented to compare the different topologies. The system configuration and different topologies are illustrated in Fig. 10. The system is composed of four wind farms rated at 1000 MW and four substations of 1000 MW connected to the main grid. The different units have been located as described in Table 2.

The analysis of the different configurations allows to extract the results summarized in Table 3, where utilization is defined as the quotient between the steady state power of a circuit in normal operation and its overall rating, N_L is the number of lines and N_{cb} is the number of HVDC circuit breakers. From the results it can be observed:

- PPT needs no circuit breakers and the lengths and line rating are acceptable
- GRT needs 8 circuit breakers with the lowest line distances but with highest line rating
- SGRT needs 8 circuit breakers and has lower line ratings
- SSRT and WFRT need only 4 circuit breakers but line length are considerably high

The flexibility of GRT and WFRT is noted specially in the event of a permanent single fault in a DC circuit close to a wind farm (with the whole system operating at maximum wind power). If there is a line fault, 1 GW from a wind farm will be lost in the cases of PPT, SGRT and SSRT. However in the cases of GRT and WFRT there is no

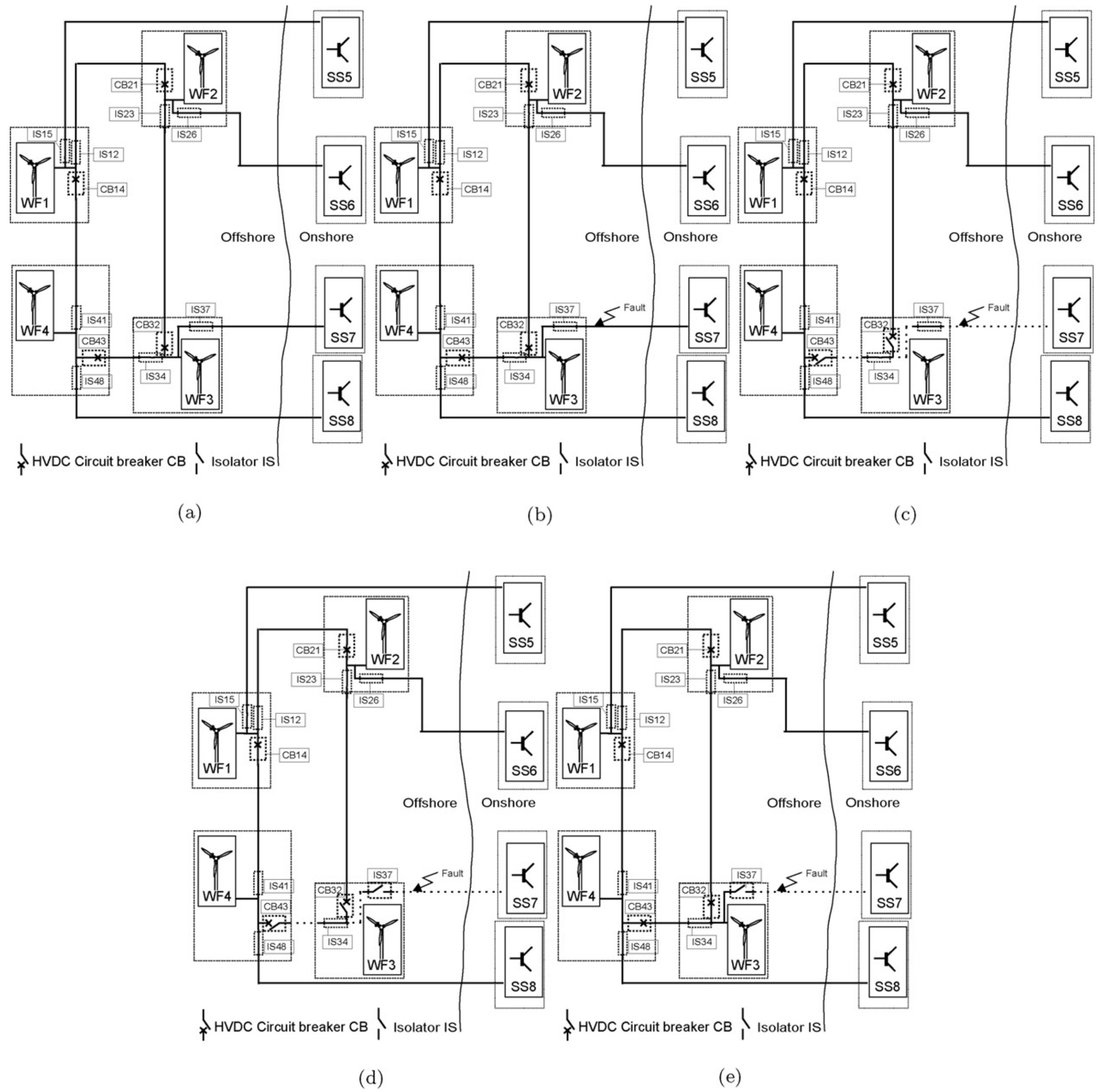

Fig. 9. Wind farm ring topology operation under a line fault. (a) Wind farm ring topology normal operation, (b) t_0 fault in the line L37, (c) t_1 circuit breakers CB32 and CB43 open, (d) t_2 isolator IS37 opens, (e) t_3 circuit breakers CB32 and CB43 close, reestablishing the connection of the wind farm WF3.

Table 1
Comparison of the proposed HVDC topologies for large offshore wind farms.

Topology		Offshore platform	Communication	Flexibility	Redundancy	Overall
PPT		No	No	No	No	Simple but it lacks flexibility
GRT		No	Yes	Good	Good	Flexible but some circuits have to be rated for the full system power
	SS side	Yes	No	Good	Yes	The circuits' rating equal to the rating of the wind
ST	WF side	Yes	No	Bad	Yes	farm or substation to which the circuit is connected,
	Total	Yes	No	Poor	Yes	but it has a weak point at the central node
SGRT						Has the advantages of both GRT and ST, but still needs full power in the central switching ring.
WFRT		No	Yes	Good	Good	Allow the isolation of a faulty circuit as in the case of point
	Ring	No	Yes	Good	Yes	to point topology and without needing
SSRT	Line	No	Yes	Bad	Yes	full system power rating of the ring
	Total	No	Yes	Poor	Yes	circuits

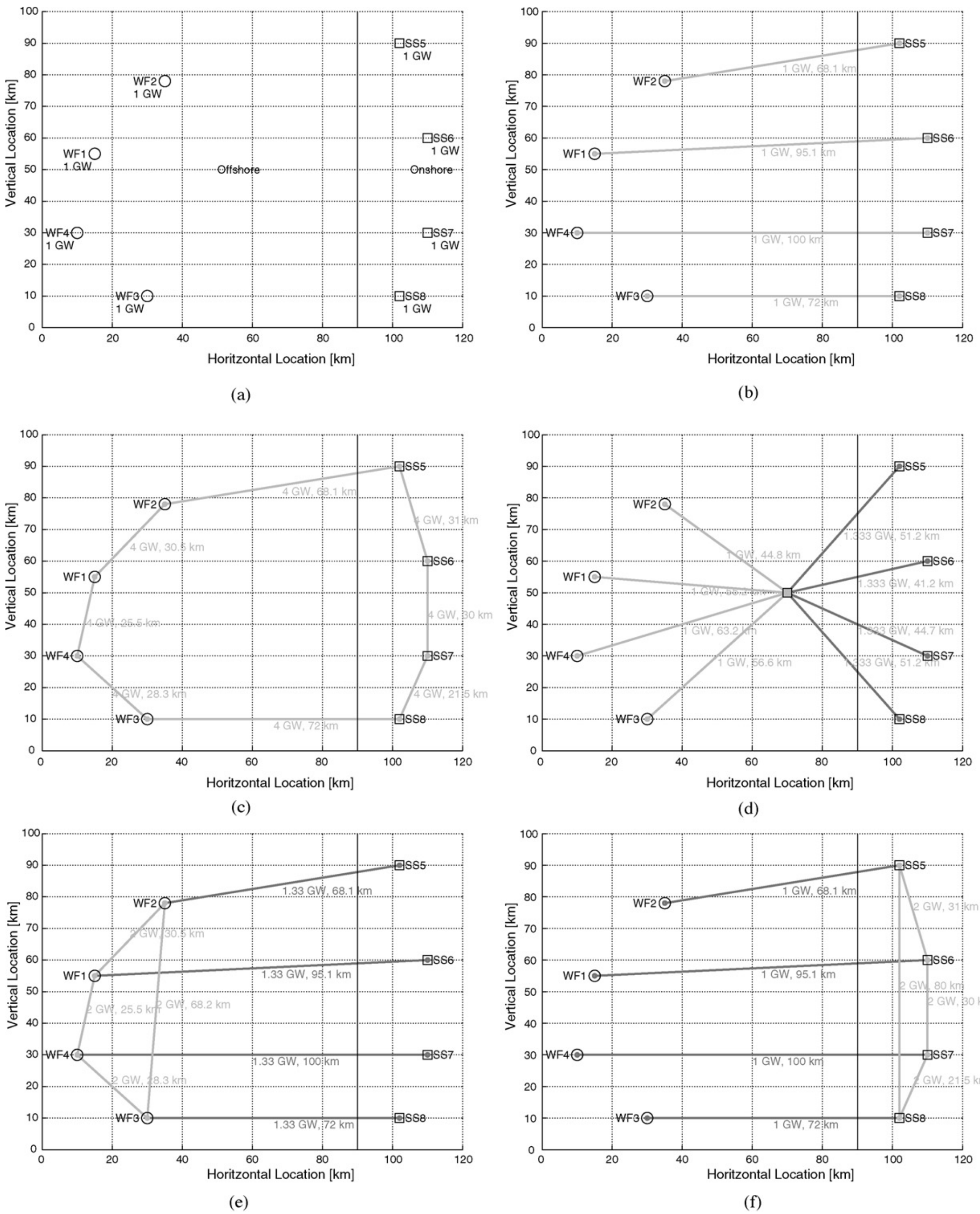

Fig. 10. Different topologies applied to the proposed example system. (a) Example system analyzed comprising four wind farms and four substations, (b) PPT topology, (c) GRT topology, (d) SGRT topology, (e) WFRT topology, and (f) SSRT topology.

power lost. In the event of a permanent fault in a DC circuit close to a substation or fault in the substation HVDC-VSC (with the whole system operating at maximum wind power), 1 GW is lost from a wind farm for PPT while there is no power loss for GRT, SGRT, WFRT and SSRT configurations.

For all the case studies, the lines with the highest rating (2–4 GW) have the lowest utilization, whereas the lines with the lowest rating (1 GW) operate at full utilization. This is due to the additional level of redundancy which the higher rated lines are providing.

Comparing the different topologies, it can be stated that WFRT topology shows good performance because it can withstand different faults without loosing power and using only four HVDC circuit breakers. If shorter circuit lengths are required, GRT can be also a good candidate but it needs some circuits rated at 4 GW.

Table 2
Location of the different wind farms and substations of the example.

WF/SS	Horizontal location (km)	Vertical location (km)
Wind Farm WF1	15	55
Wind Farm WF2	35	78
Wind Farm WF3	30	10
Wind Farm WF4	10	30
Substation SS5	102	90
Substation SS6	110	60
Substation SS7	110	30
Substation SS8	102	10

Table 3
Case study results.

Topology		N_L	Lines length (km)	Lines rating (GW)	N_{cb}	Utilization
PPT		4	335	1	0	100%
GRT		8	307	2–4 GW	8	25–50%
	SS side	4	220	1.33	4	75%
SGRT	WF side	4	188	1	4	100%
	Total	8	408		8	
	ring	4	152	2	4	50%
WFRT	line	4	335	1.33	0	75%
	Total	8	488		4	
	ring	4	163	2	4	50%
SSRT	line	4	335	1	0	100%
	Total	8	498		4	

6. Conclusions

Multiterminal HVDC-VSC transmission systems for large offshore wind farms have been considered, focusing on various control approaches and HVDC circuit topologies. Different HVDC grid topologies have been studied considering system faults. The topologies presented include Point to point topology, General ring topology, Star topology, Star with a central switching ring topology, Wind farms ring topology, and Substation ring topology. Each topology has been characterised in terms of number and capacity of circuits needed, number of circuit breakers required, the possibility of meeting the maximum loss criterion, flexibility, redundancy, lines utilization, necessity of an offshore platform and requirements of communications.

The analysis performed shows that selecting the optimum topology for a given application depends not only on operation and robustness requirements but also on the geographical location of the substations and wind farms and the eventual cost of HVDC circuit breakers and cables. With the current high cost of solid-state HVDC circuit breakers, it seems that the most appropriate configuration would be the WFRT, since it can meet all the required criteria with a reduced number of solid-state HVDC circuit breakers and without needing an offshore switching platform. However, detailed analysis are required before selecting the best topology for a given application as factors such as distances between nodes, cable and circuit breaker costs and communication system availability, will vary from project to project.

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